Jacob Bauman

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Education

University of Pennsylvania — Exp. Grad: Dec 2027
Master of Science in Robotics — *Philadelphia, PA*

- **Specialization**: Machine Learning and AI
- **Current Coursework**: Autonomous Car Racing (F1TENTH), Controls & Optimization for Robotics, Applied ML
- **Planned (Spring 2027)**: Robot Learning, Physical Intelligence, Learning and Control for Robotics

University of California, Santa Barbara — June 2026
Bachelor of Science in Mechanical Engineering with Honors, GPA: 3.91 — *Santa Barbara, CA*

- **Specialization**: Robotics and Control

Research & Projects

Controls & Perception Lead | *Autonomous Car Racing (F1TENTH)* **[watch videos]** — Aug 2026 – Present

- Developed **MPC path-tracking controller**; validated in sim and deployed on autonomous race car, **placing 4th of 14 teams** in obstacle race with **2-person team vs 4-person teams**
- Implemented an **extended Kalman filter** (wheel odometry + IMU + dynamics) feeding **particle-filter** localization on hardware; mapped the track with LiDAR **SLAM** to ∼7 cm localization error at race speed
- Implemented minimum-curvature raceline **trajectory planner** as a QP; with MPC, cut lap time 39% in sim and 26% on hardware (16.4 → 12.1s)
- Calibrated car hardware (steering, actuation delay, sensor offset, VESC), cut **tracking error 80%** (21.6 → 4.2 cm)
- Designed LiDAR **obstacle detection and avoidance planner** as a ROS2 node feeding the MPC

Co-lead Robotics Researcher | *Transformable Leg-Wheel Robot* — Jan 2026 – Present

- Developing **reinforcement learning policy** that adapts leg extension and wheel torques to terrain, using a base policy with an online adaptation module
- Built GPU-accelerated MuJoCo simulation of robot and terrain to train RL control policies
- Deploying policy on hardware, using domain randomization and system identification to close the sim-to-real gap
- **First-author IROS paper** in preparation for March 2027 submission deadline

Controls Lead | *Robust Adaptive MPC Quadruped Locomotion* — Oct 2026 – Present

- Building a hierarchical controller for goal-directed locomotion with (trajectory planner → gait/foothold planner → centroidal MPC → joint torque control)
- Integrating adaptive robust MPC with a direct collocation global planner and control barrier function filter into an open-source quadruped control stack (Quadruped-PyMPC, acados, MuJoCo) for a simulated Unitree Go2

Drone (UAV) Project Lead | *Autonomous Vehicle Racing Capstone Project* — Sept 2025 – June 2026

- Led 5-person team to **1st place** in Raytheon Autonomous Vehicle Competition
- **Developed GPS-denied perception and control** to search for, track, and land on a moving ArUco-tagged platform in 10 mph winds

Project Manager | *Tabletop Swarm Robots, 4 person team* — March 2025 – Oct 2025

- Designed three robot system to collaboratively and autonomously pick up, transport, and place plates from random start positions to target location
- **Won best-in-class project** among 23 teams in the design course
- Implemented multi-robot A* to compute optimal collision-free paths accounting for robot dynamics
- Implemented real-time, multi-agent plate stabilization, crash-detection, and autonomous evasive maneuvers

Industry Experience

Controls & Software Engineering Intern — June 2026 – Aug 2026
Lawrence Livermore National Laboratory - Adaptive Optics Group — *San Francisco Bay Area, CA*

- Developed and deployed a closed-loop star-tracking control system for a research telescope, holding pointing accuracy within ±0.002° over a full night (**in active use** by researchers)
- Built a camera-based perception pipeline to detect and identify stars in real time at 20 Hz, integrated with telescope mount and actuators

- Developed simulator generating synthetic star-field images to tune and validate the detection algorithm, improving detection rate by 41% across environmental conditions

Controls Engineering Intern June 2025 – Sept 2025
Lawrence Livermore National Laboratory - Adaptive Optics Group *San Francisco Bay Area, CA*

- Designed and deployed **automated laser alignment system** on operational research telescope
- Built control pipeline with proportional **closed-loop controller** and **self-calibrating** control matrix
- **Designed and validated state estimation algorithms** using **custom simulation** and benchmarked against statistically ideal performance

Mechanical Engineering Intern June 2024 – June 2025
Lawrence Livermore National Laboratory - Space Program *San Francisco Bay Area, CA*

- **Automated** optical **testing** with closed-loop Python program to characterize optical errors and automate focus correction using signal/image processing, Fourier analysis, and statistical error analysis
- **Increased image acquisition speed by 100×** and **improved precision ∼40×** (to 1 pixel, 2.4 µm) over previous methods by automating the image data collection system

Technical Skills

Controls, Planning & Learning: MPC, LQR, PID, Trajectory Optimization, IK, A*, RRT, RL
Perception & State Estimation: SLAM, System ID, Particle Filters, EKF, Sensor Fusion, OpenCV, Optical Flow
Software: ROS2, Python, C++, Git, Linux, JAX, MuJoCo, Gazebo, RViz, Foxglove, convex optimization, Acados
Hardware: Jetson Orin, Raspberry Pi, Pixhawk, Arduino, BLDC Motors, VESC, Quadrotors, LIDAR, IMU